1. Two players are gambling with an unfair coin (probability of head 0.3). Initially, Player 1 has \$3 and Player 2 has \$2. After each toss, Player 1 wins \$1 if head occurs and loses \$1 if tail occurs. The game ends when one player has all the money. Let X_n be the amount of money Player 1 has at "time" n. Calculate $P(X_3 = 2)$.

2. Use the following data to estimate the transition probabilities and construct the Transition Probability Matrix.

   1 10101 01011 10111 01011 11100 11110 10011 10110 10101

3. Let $S = \{1, 2, 3\}$ and $P = \begin{pmatrix} 0.2 & 0.1 & 0.7 \\ 0.3 & 0.3 & 0.4 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$. Calculate $P(X_2 = 2 \mid X_0 = 3)$.

4. There are two brands, I and II, in the market. Out of those who buy brand I in one month, 30% switch to brand II during the next month. Out of those who buy brand II in one month, 20% switch to brand I during the next month. Construct the transition probability matrix, determine the equilibrium distribution and comment.

5. There are three brands in the market: 1, 2 and 3. Customer preferences are observed at the beginning of each month. The initial distribution of customer preference is $\underline{a}^{(0)} = (0.3 \;\; 0.3 \;\; 0.4)$. The TPM is:

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

   Calculate the marginal distributions of X_1 and X_2.

6. Check whether the Markov Chain with the following transition probability matrix is irreducible. Classify its states and determine the periods.

$$P = \begin{pmatrix} 0.4 & 0 & 0.6 & 0 \\ 0 & 0.7 & 0 & 0.3 \\ 0 & 0.2 & 0 & 0.8 \\ 0.5 & 0 & 0.5 & 0 \end{pmatrix}$$

7. Check whether the Markov Chain with the following transition probability matrix is irreducible. Classify its states and determine the periods.

$$P = \begin{pmatrix} 0 & 0.3 & 0 & 0.4 & 0.3 & 0 \\ 0 & 0.3 & 0 & 0.3 & 0.4 & 0 \\ 0.1 & 0.2 & 0.1 & 0.2 & 0.1 & 0.1 \\ 0 & 0.2 & 0 & 0.2 & 0.6 & 0 \\ 0 & 0.6 & 0 & 0.2 & 0.2 & 0 \\ 0.1 & 0 & 0.2 & 0.2 & 0.2 & 0.3 \end{pmatrix}$$

8. John starts with \$2 and $p = 0.6$. What is the probability that John wins if the opponent also has \$5?

9. John starts with \$3 and $p = 0.5$. What is the probability that John wins if the opponent also has \$5?

10. John starts with \$2 and $p = 0.6$. What is the probability that John will become infinitely rich if the opponent has infinite capital?

11. John starts with \$2 and $p = 0.5$. What is the probability that John will become infinitely rich if the opponent has infinite capital?

12. You bought a share of stock for \$12. Stock price moves \$1 each day as a simple random walk. The probability of an increase is 0.52.
    a. What is the probability that the stock price will reach \$16 before reaching \$9?
    b. What is the probability that you will get infinitely rich?

13. Customers arrive at a bank according to a Poisson process with rate $\lambda = 6/\text{hour}$. Determine
    a. the expected number of customers in 7 hours.
    b. the probability that nobody arrives from 2:00 to 2:15 p.m.
    c. the probability that 2 customers arrive from 2:00 to 2:15 p.m.
    d. the average waiting time between two successive arrivals.

14. A vendor sells fish kababs as follows. Every day he goes to a new place and gives fish kababs to three customers for free. From then on, each customer brings new customers with a rate of $0.9/\text{hour}$.
    a. What is the probability that after 3 hours the total number of customers will be 7?
    b. What is the average number of customers during a 2-hour period?

15. In a store, there are 12 packets of candy for sale. The sale rate depends on the number of packets remaining. For one packet, the rate is $0.6/\text{hour}$.
    a. What is the probability that after 2 hours 7 packets will be remaining?
    b. What is the probability that all the packets will be sold out in an 8-hour day?
    c. What is the average number of items sold in 4 hours?

16. Arrival at a telephone booth forms a Poisson process with mean $10/\text{hour}$. Length of a telephone call follows exponential with mean 2 minutes.

a. What is the probability that, on arrival, a person will find the telephone occupied?
b. What is the average length of queue?
c. What is the average waiting time?
d. What is the probability that a newcomer has to stay in the system for more than 3 minutes?

17. A supermarket has 3 girls ringing up sales at the counters. People arrive in a Poisson process with a rate of 15/hour. Service time follows exponential distribution with mean 4 minutes.
    a. Under steady state condition, what is the probability that all the girls will be busy?
    b. What is the average number of customers in the system?
    c. What is the average waiting time of a customer who has just arrived?